

Elastic Security SIEM & Automated Incident Response Project

By: Ritvik Indupuri

Executive Summary

Developed an end-to-end security monitoring and automated incident response pipeline using Elastic SIEM and Tines SOAR. This project showcases real-time detection of administrative login events, automated incident triage, and immediate alerting workflows, simulating enterprise SOC operations.

Objective

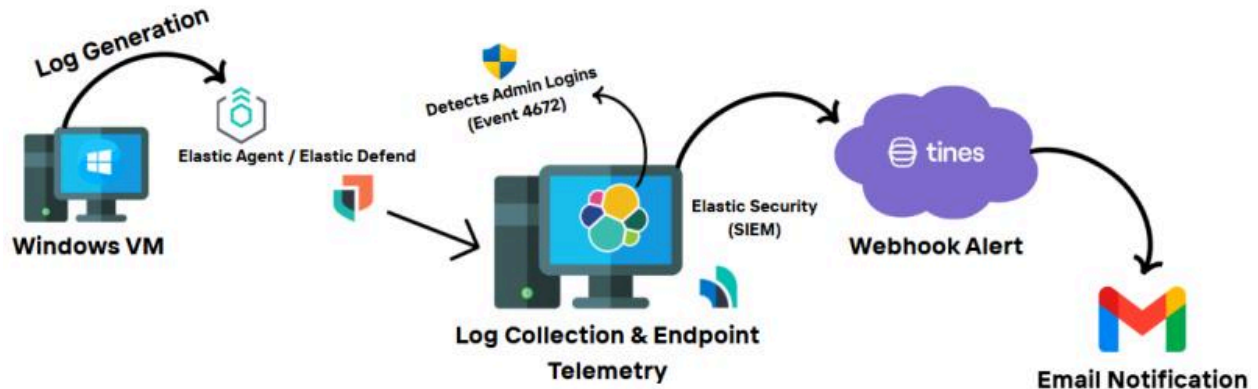
Designed and implemented a full-stack security monitoring and automated incident response system using **Elastic Security (SIEM/EDR)** and **Tines (SOAR)**. This project focused on real-time log ingestion, threat detection, and automated alert notifications to streamline incident response workflows. The goal was to simulate enterprise-grade SOC operations, enhancing visibility into endpoint activity and reducing response latency.

Environment Overview

- **Log Source:** Windows 10 Virtual Machine
 - **Security Agent:** Elastic Agent with Elastic Defend (EDR)
 - **SIEM Platform:** Elastic Security (Kibana/Elasticsearch Stack)
 - **SOAR Integration:** Tines (Webhook Automation)
 - **Notification Channel:** Automated Email Alerts via Tines Workflow
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System Architecture

Figure 1: End-to-End Workflow Architecture



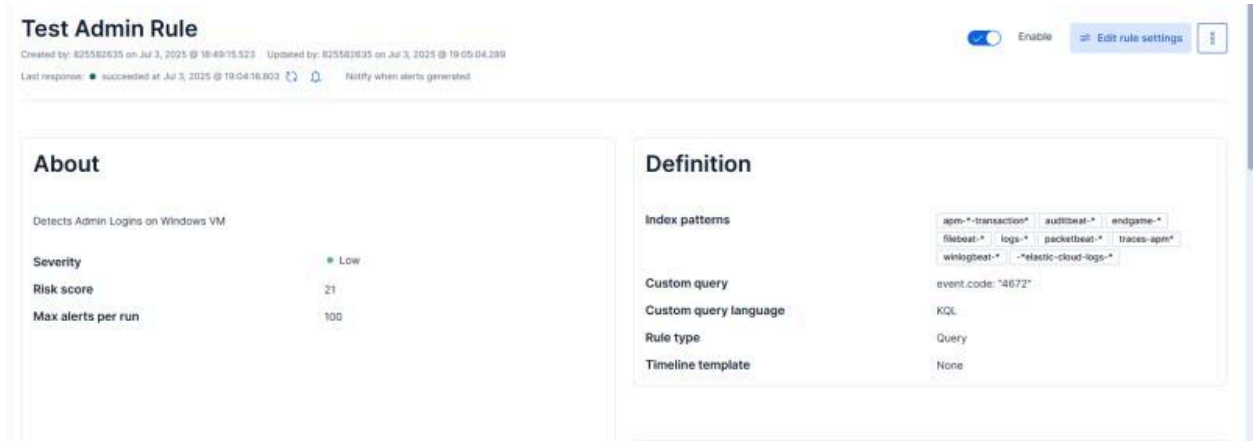
Description:

The architecture diagram illustrates the flow of telemetry from log generation to incident notification:

1. Windows VM generates security events.
 2. Elastic Agent (Defend) collects logs and endpoint telemetry.
 3. Events are ingested into Elastic Security (SIEM) for correlation and detection.
 4. Upon detection of admin login events, alerts trigger a Tines webhook.
 5. Tines summarizes incident details using AI and dispatches automated email notifications.
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Detection Rule Implementation

Figure 2: Custom SIEM Rule — Test Admin Rule



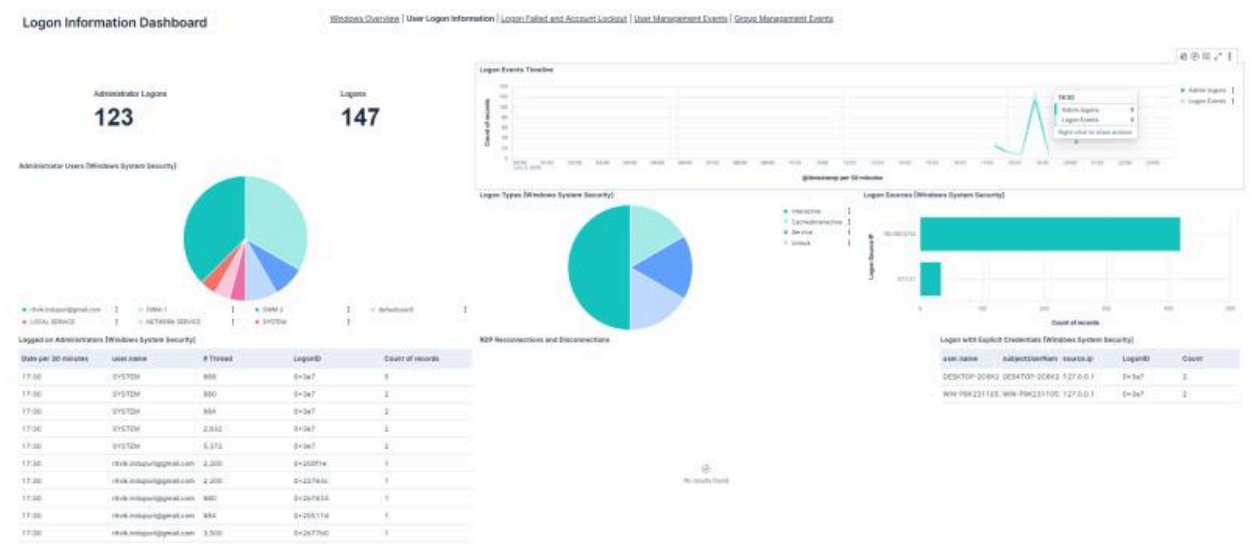
Description:

A custom SIEM rule was created to detect administrative login events on the Windows VM by querying for **event.code: "4672"**. The rule was configured with a severity level of Low, a risk score of 21, and a maximum alert threshold of 100 per run.

- **Rule Type:** KQL Query
 - **Trigger Condition:** Admin logon detection (event code 4672)
 - **Response Action:** Send webhook to Tines upon alert generation
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Alert Visualization & Dashboarding

Figure 3: Logon Information Dashboard

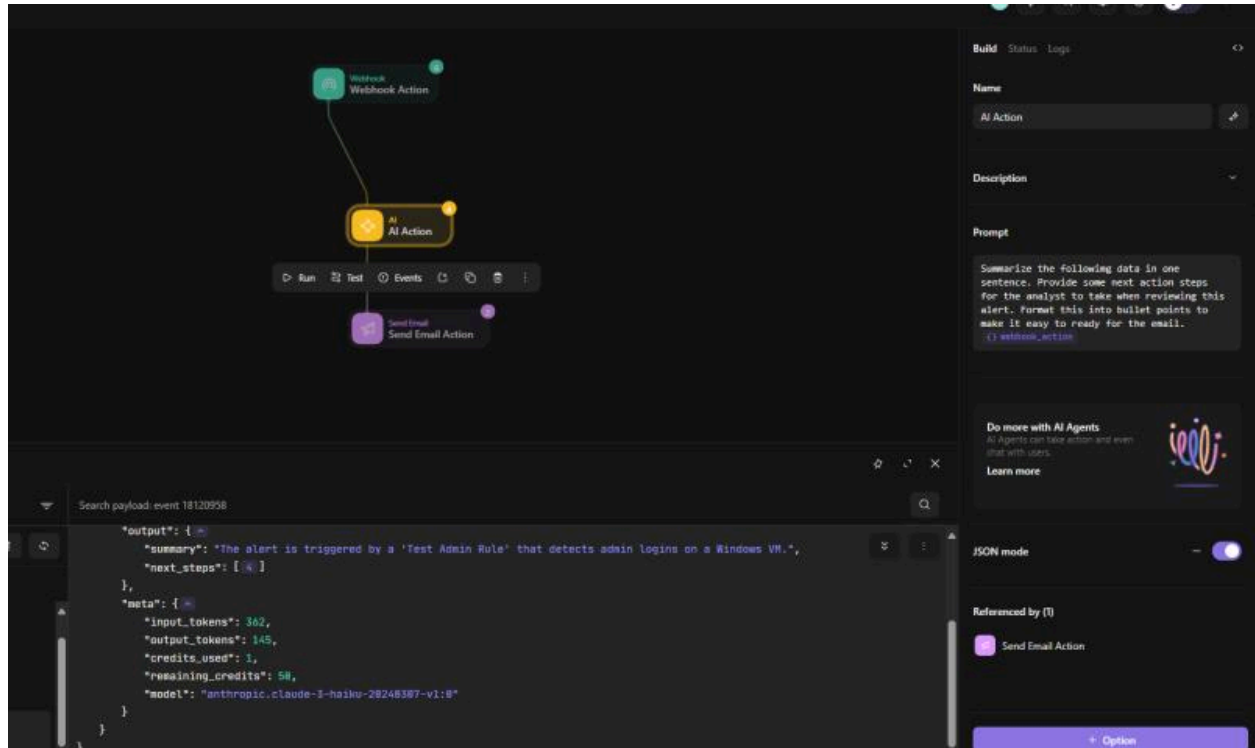


Description:

The logon information dashboard visualizes administrator login activities, RDP sessions, logon types, and event timelines. Pie charts and graphs provide a quick overview of logon patterns, enabling analysts to identify abnormal spikes or unauthorized access attempts in real-time.

Automated Response Workflow

Figure 4: Tines SOAR Workflow Design



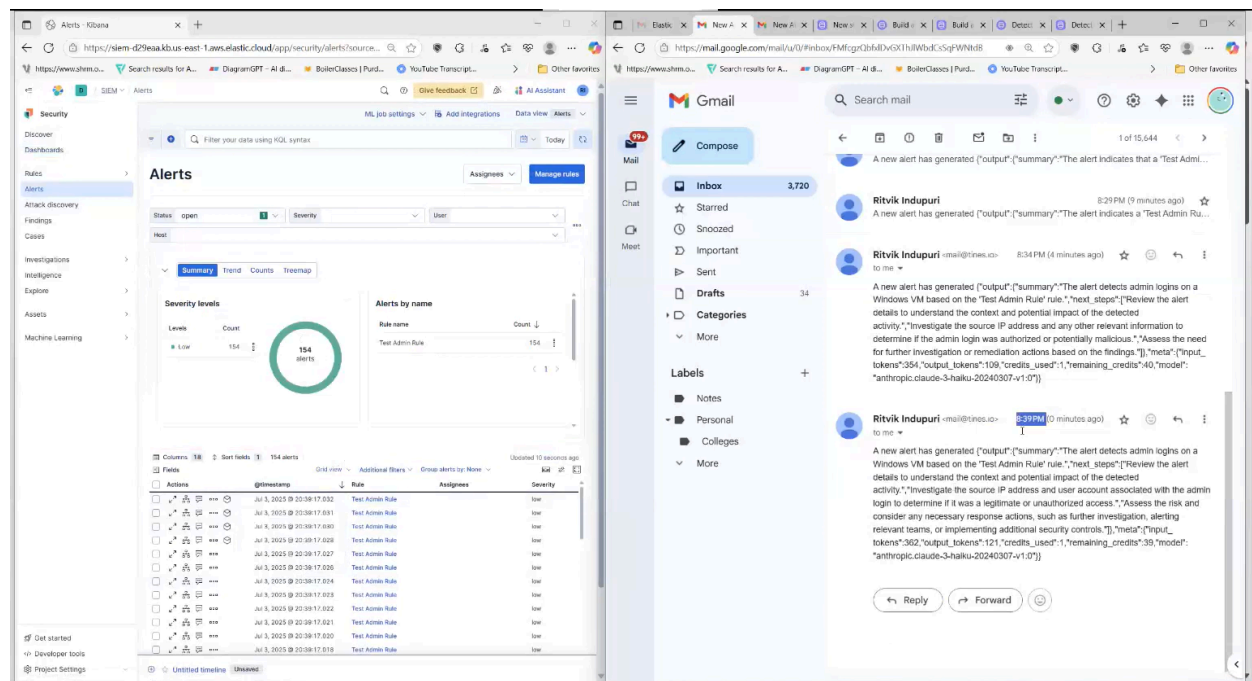
Description:

Tines orchestrates the incident response process by:

1. Receiving webhook alerts from Elastic SIEM.
2. Processing the alert payload through an AI summarization action.
3. Sending structured email notifications with incident details to SOC teams for immediate awareness.

Real-time Alerting & Email Notifications

Figure 5: Alerts and Email Notification Snapshot



Description:

The Elastic SIEM dashboard displays active alerts generated by the custom detection rule, while the Gmail interface shows corresponding automated alert emails. This real-time notification pipeline ensures rapid incident awareness and response.

Impact and Conclusion

This project demonstrates a fully integrated SIEM and SOAR pipeline designed for real-time threat detection and automated incident response. By automating the initial triage of 154 administrative login alerts, the system reduces manual analyst workload and accelerates incident awareness, simulating an enterprise SOC workflow. By configuring custom detection rules in Elastic Security and orchestrating automated alert triage using Tines, the system mirrors modern enterprise SOC workflows. The result is a scalable, efficient model for reducing alert fatigue and accelerating incident handling in security operations.